

Roles and Responsibilities of On-Prem SQL Server DBA, AWS Cloud DBA, and Azure Cloud DBA

Area	On-Prem SQL Server DBA	AWS SQL Server DBA	Azure SQL Server DBA
Infrastructure Management	Responsible for physical server setup, storage configuration, OS patching, and hardware maintenance.	No physical hardware; responsible for provisioning EC2 instances (for self-managed SQL Server) or using Amazon RDS/SQL Server with managed infrastructure.	No physical hardware; responsible for provisioning Azure SQL Database, Managed Instance, or SQL on VM. Infrastructure is mostly managed by Azure.
Installation & Upgrades	Full installation of SQL Server, applying service packs, and version upgrades manually.	Install SQL Server on EC2 or configure RDS; RDS handles patching/upgrades automatically if enabled.	Install SQL Server on Azure VM or provision Azure SQL PaaS services; Azure handles patching for PaaS.
Backup & Recovery	Manual backup strategies using native SQL backups, custom scripts, and storage systems.	Configure RDS automated backups or manage backups for EC2 SQL Server; S3 often used for storage.	Configure automated backups for Azure SQL Database/Managed Instance or manage backups for SQL on VM; backups stored in Azure Blob Storage.
High Availability (HA) & Disaster Recovery (DR)	Configure Always On Availability Groups, Failover Clustering, Log Shipping, Mirroring manually.	Use RDS Multi-AZ for HA or manual Always On for EC2 deployments.	Use Azure SQL Auto-failover groups, Geo-replication, or manual Always On for SQL on VM.
Security & Compliance	Configure firewalls, encryption, user roles, and auditing manually.	Configure AWS Security Groups, IAM roles, encryption (KMS), and RDS security settings.	Configure NSGs, Azure AD integration, Transparent Data Encryption (TDE), and role-based access.
Performance Monitoring & Tuning	Use native tools (Profiler, Extended Events, DMVs) and custom monitoring systems.	Use CloudWatch metrics, Performance Insights for RDS, and DMVs for tuning.	Use Azure Monitor, Query Performance Insights, and DMVs for tuning.
Patching	Download and apply patches manually; schedule downtime.	AWS RDS can auto-patch; EC2 SQL Server needs manual patching.	Azure SQL PaaS auto-patches; Azure VM SQL Server needs manual patching.
Cost Management	Manage hardware purchase, licensing, and maintenance costs.	Manage AWS resource usage, instance sizing, storage, and reserved instance planning.	Manage Azure resource usage, DTU/vCore sizing, storage, and reserved capacity planning.
Automation & Scripting	Primarily T-SQL, PowerShell scripts, SQL Agent jobs.	AWS CLI, PowerShell, SDK, and automation with AWS Lambda for DB tasks.	Azure CLI, PowerShell, ARM templates, and automation with Azure Automation Runbooks.
Scalability	Limited by physical hardware; scaling often means buying/upgrading hardware.	Scale vertically (instance type) or horizontally (read replicas) easily.	Scale up/down service tiers or add replicas with minimal downtime.
Disaster Recovery Testing	Fully manual DR drill process.	Can test by promoting read replicas or using cross-region RDS snapshots.	Can test by failing over to Geo-replicated database or secondary region.

Visual chart showing where responsibilities decrease, stay same, or increase when moving from On-Prem → AWS → Azure for a DBA.

